Depreciation

Exam code: 0607

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Depreciation

What does depreciation mean?

- Depreciation is where an item **loses value** over time
  - E.g. cars, mobile phones, etc
- Depreciation is usually calculated as a percentage decrease at the end of each year
  - This works the same as compound interest, but with a percentage **decrease**

How do I calculate depreciation?

- A similar method to **compound interest** can be used
- Change the **multiplier** to one which represents a **percentage decrease**
  - e.g. a **decrease of 15%** would be a **multiplier** of **0.85**
- If a car worth \$16 000 depreciates by 15% each year for 6 years
  - Its value will be $16\,000 \times 0.85^6$, which is \$6034.39
- If you are asked to find the amount the value has depreciated by:
  - Find the difference between the starting value and the new value

Depreciation formula

- An **alternate method** is to use the **following formula** to calculate the final balance
  - Final balance = $P\left(1 - \frac{r}{100}\right)^n$ where
    - P is the original amount,
    - r is the % increase
    - and n is the number of years
  - Note that all of $1 - \frac{r}{100}$ is the **multiplier**
    - e.g. 0.75 for a 25% depreciation
- This formula is **not given** in the exam

Help me understand

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Got it

by 15%.

Find the value of the car after 3 full years.

Answer:

Identify the multiplier

$$100\% - 15\% = 85\%$$

$$m = 1 - 0.15 = 0.85$$

Raise to the power of number of years

$$0.85^3$$

Multiply by the starting value

$$£20\,000 \times 0.85^3$$

£12 282.50

Alternative method

Use the formula for the final amount $P\left(1 - \frac{r}{100}\right)^n$

Substitute $P = 20\,000$, $r = 15$ and $n = 3$ into the formula

$$20\,000\left(1 - \frac{15}{100}\right)^3$$

£12 282.50

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